



## 1. Description

### 1.1. Project

|                 |                     |
|-----------------|---------------------|
| Project Name    | holding_cabinet_two |
| Board Name      | NUCLEO-L476RG       |
| Generated with: | STM32CubeMX 6.18.1  |
| Date            | 10/06/2026          |

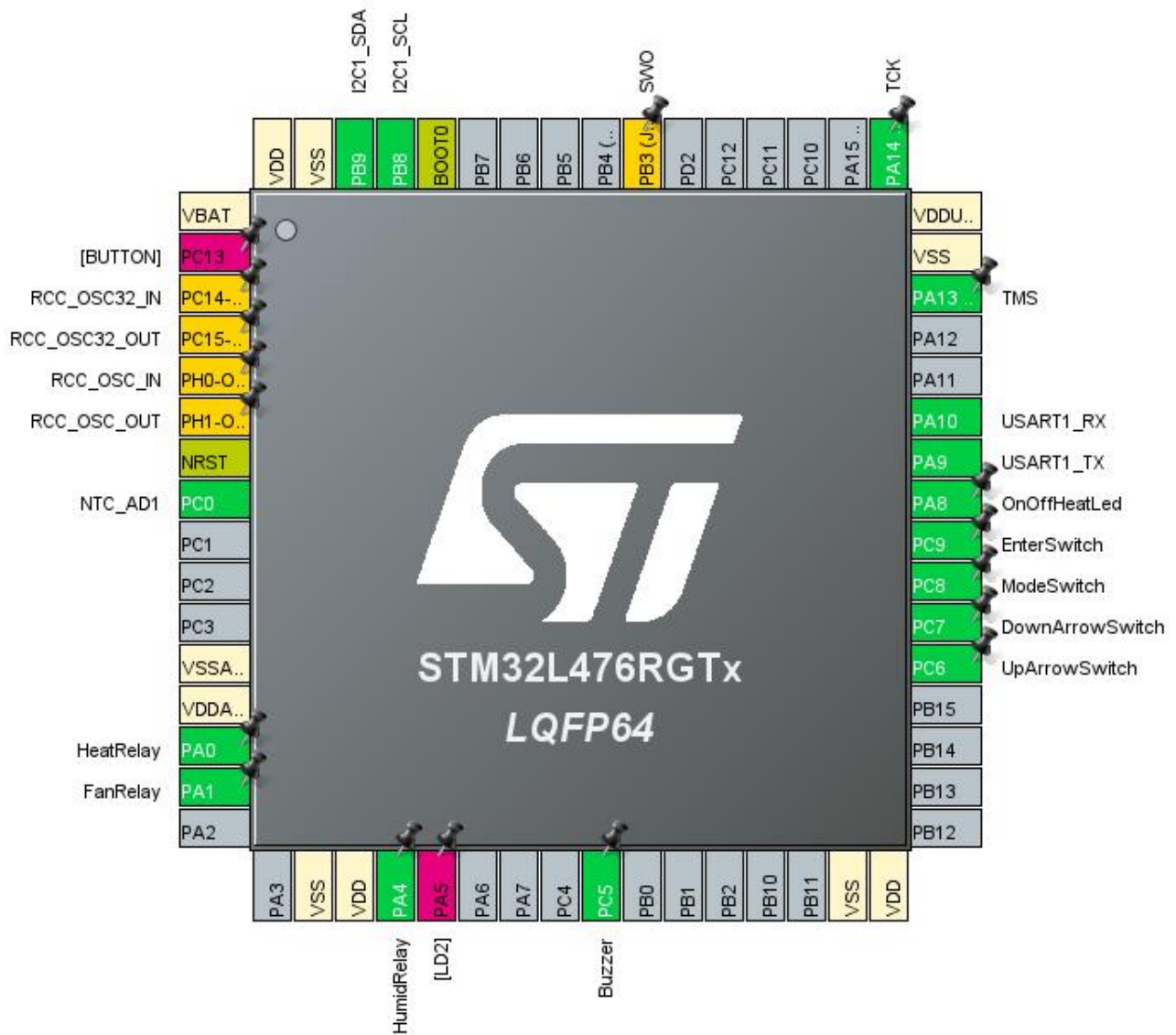
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32L4       |
| MCU Line       | STM32L4x6     |
| MCU name       | STM32L476RGTx |
| MCU Package    | LQFP64        |
| MCU Pin number | 64            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration



### 3. Pins Configuration

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label           |
|----------------------|---------------------------------------|----------|--------------------------|-----------------|
| 1                    | VBAT                                  | Power    |                          |                 |
| 2                    | PC13                                  | I/O      |                          |                 |
| 3                    | PC14-OSC32_IN (PC14) *                | I/O      | RCC_OSC32_IN             |                 |
| 4                    | PC15-OSC32_OUT (PC15)<br>*            | I/O      | RCC_OSC32_OUT            |                 |
| 5                    | PH0-OSC_IN (PH0) *                    | I/O      | RCC_OSC_IN               |                 |
| 6                    | PH1-OSC_OUT (PH1) *                   | I/O      | RCC_OSC_OUT              |                 |
| 7                    | NRST                                  | Reset    |                          |                 |
| 8                    | PC0                                   | I/O      | ADC1_IN1                 | NTC_AD1         |
| 12                   | VSSA/VREF-                            | Power    |                          |                 |
| 13                   | VDDA/VREF+                            | Power    |                          |                 |
| 14                   | PA0 **                                | I/O      | GPIO_Output              | HeatRelay       |
| 15                   | PA1 **                                | I/O      | GPIO_Output              | FanRelay        |
| 18                   | VSS                                   | Power    |                          |                 |
| 19                   | VDD                                   | Power    |                          |                 |
| 20                   | PA4 **                                | I/O      | GPIO_Output              | HumidRelay      |
| 21                   | PA5                                   | I/O      |                          |                 |
| 25                   | PC5 **                                | I/O      | GPIO_Output              | Buzzer          |
| 31                   | VSS                                   | Power    |                          |                 |
| 32                   | VDD                                   | Power    |                          |                 |
| 37                   | PC6 **                                | I/O      | GPIO_Input               | UpArrowSwitch   |
| 38                   | PC7 **                                | I/O      | GPIO_Input               | DownArrowSwitch |
| 39                   | PC8 **                                | I/O      | GPIO_Input               | ModeSwitch      |
| 40                   | PC9 **                                | I/O      | GPIO_Input               | EnterSwitch     |
| 41                   | PA8 **                                | I/O      | GPIO_Output              | OnOffHeatLed    |
| 42                   | PA9                                   | I/O      | USART1_TX                |                 |
| 43                   | PA10                                  | I/O      | USART1_RX                |                 |
| 46                   | PA13 (JTMS-SWDIO)                     | I/O      | SYS_JTMS-SWDIO           | TMS             |
| 47                   | VSS                                   | Power    |                          |                 |
| 48                   | VDDUSB                                | Power    |                          |                 |
| 49                   | PA14 (JTCK-SWCLK)                     | I/O      | SYS_JTCK-SWCLK           | TCK             |
| 55                   | PB3 (JTDO-TRACESWO) *                 | I/O      | SYS_JTDO-SWO             | SWO             |
| 60                   | BOOT0                                 | Boot     |                          |                 |
| 61                   | PB8                                   | I/O      | I2C1_SCL                 |                 |
| 62                   | PB9                                   | I/O      | I2C1_SDA                 |                 |
| 63                   | VSS                                   | Power    |                          |                 |

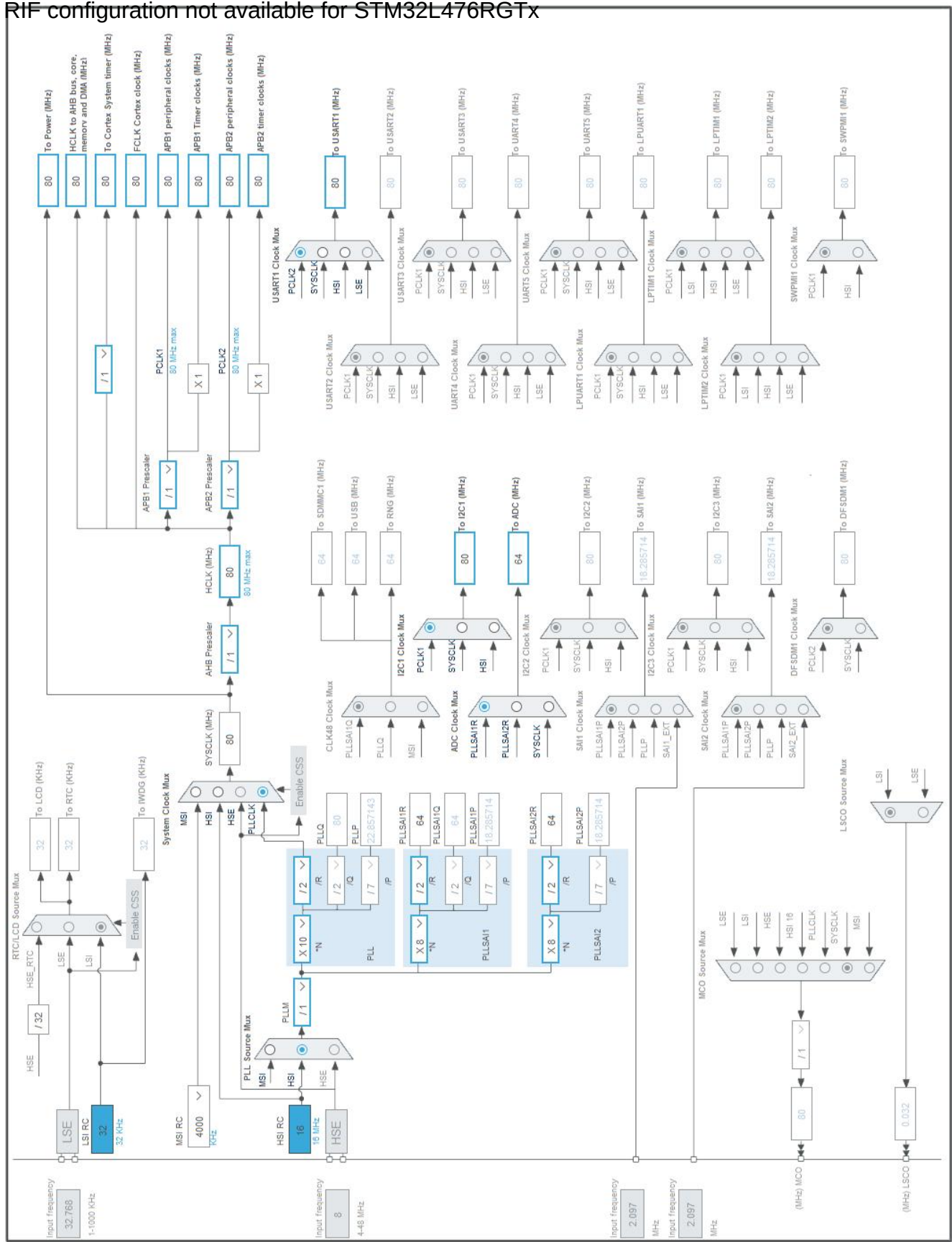
| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|----------------------|---------------------------------------|----------|--------------------------|-------|
| 64                   | VDD                                   | Power    |                          |       |

\*\* The pin is affected with an I/O function

\* The pin is affected with a peripheral function but no peripheral mode is activated

## 4. Clock Tree Configuration

~~RIF configuration not available for STM32L476RGTx~~



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32L4       |
| Line      | STM32L4x6     |
| MCU       | STM32L476RGTx |
| Datasheet | DS10198 Rev4  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

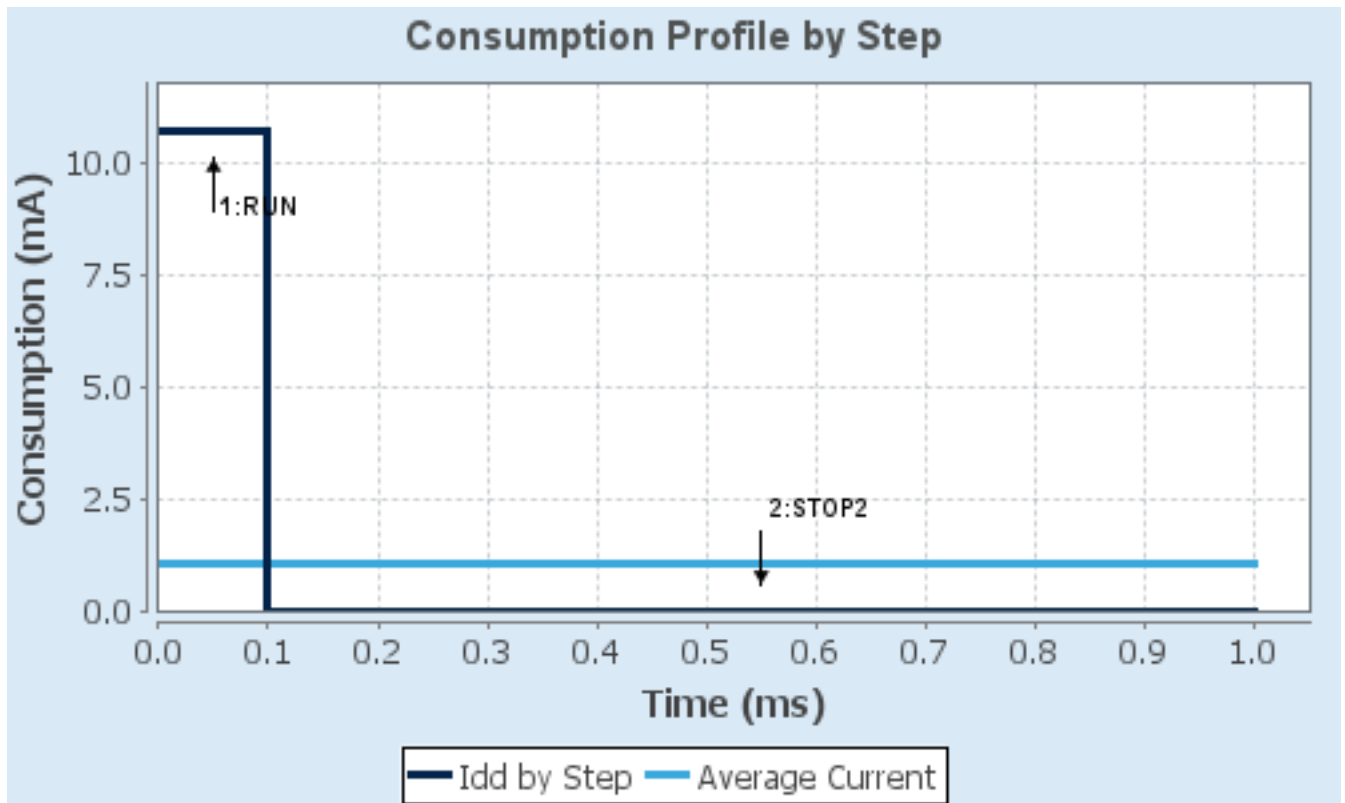
#### 1.4. Sequence

|                               |             |                |
|-------------------------------|-------------|----------------|
| <b>Step</b>                   | Step1       | Step2          |
| <b>Mode</b>                   | RUN         | STOP2          |
| <b>Vdd</b>                    | 3.0         | 3.0            |
| <b>Voltage Source</b>         | Battery     | Battery        |
| <b>Range</b>                  | Range1-High | NoRange        |
| <b>Fetch Type</b>             | SRAM2       | n/a            |
| <b>CPU Frequency</b>          | 80 MHz      | 0 Hz           |
| <b>Clock Configuration</b>    | HSE PLL     | ALL CLOCKS OFF |
| <b>Clock Source Frequency</b> | 4 MHz       | 0 Hz           |
| <b>Peripherals</b>            |             |                |
| <b>Additional Cons.</b>       | 0 mA        | 0 mA           |
| <b>Average Current</b>        | 10.7 mA     | 1.18 $\mu$ A   |
| <b>Duration</b>               | 0.1 ms      | 0.9 ms         |
| <b>DMIPS</b>                  | 100.0       | 0.0            |
| <b>Ta Max</b>                 | 103.56      | 105            |
| <b>Category</b>               | In DS Table | In DS Table    |

#### 1.5. Results

|               |                            |                 |             |
|---------------|----------------------------|-----------------|-------------|
| Sequence Time | 1 ms                       | Average Current | 1.07 mA     |
| Battery Life  | 4 months, 10 days, 3 hours | Average DMIPS   | 100.0 DMIPS |

#### 1.6. Chart



## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                                                  |
|-----------------------------------|------------------------------------------------------------------------|
| Project Name                      | holding_cabinet_two                                                    |
| Project Folder                    | C:\Users\Bobby Taylor\OneDrive\Documents\Developer\holding_cabinet_two |
| Toolchain / IDE                   | CMake                                                                  |
| Firmware Package Name and Version | STM32Cube FW_L4 V1.18.2                                                |
| Application Structure             | Advanced                                                               |
| Generate Under Root               | No                                                                     |
| Do not generate the main()        | No                                                                     |
| Minimum Heap Size                 | 0x200                                                                  |
| Minimum Stack Size                | 0x400                                                                  |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name       | Peripheral Instance Name |
|------|---------------------|--------------------------|
| 1    | SystemClock_Config  | RCC                      |
| 2    | MX_GPIO_Init        | GPIO                     |
| 3    | MX_ADC1_Init        | ADC1                     |
| 4    | MX_I2C1_Init        | I2C1                     |
| 5    | MX_USART1_UART_Init | USART1                   |

## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

#### IN1: IN1 Single-ended

##### 3.1.1. Parameter Settings:

###### **ADCs\_Common\_Settings:**

Mode Independent mode

###### **ADC\_Settings:**

Clock Prescaler Asynchronous clock mode divided by 1

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

End Of Conversion Selection End of single conversion

Overrun behaviour Overrun data preserved

Low Power Auto Wait Disabled

###### **ADC\_Regular\_ConversionMode:**

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 1

Sampling Time **247.5 Cycles \***

Offset Number No offset

###### **ADC\_Injected\_ConversionMode:**

Enable Injected Conversions Disable

###### **Analog Watchdog 1:**

Enable Analog WatchDog1 Mode false

###### **Analog Watchdog 2:**

Enable Analog WatchDog2 Mode false

###### **Analog Watchdog 3:**

Enable Analog WatchDog3 Mode false

### 3.2. I2C1

#### I2C: I2C

##### 3.2.1. Parameter Settings:

###### Timing configuration:

|                               |                     |
|-------------------------------|---------------------|
| Custom Timing                 | Disabled            |
| I2C Speed Mode                | Standard Mode       |
| I2C Speed Frequency (KHz)     | 100                 |
| Rise Time (ns)                | 100                 |
| Fall Time (ns)                | 100                 |
| Coefficient of Digital Filter | 0                   |
| Analog Filter                 | Enabled             |
| Timing                        | <b>0x10D19CE4 *</b> |

###### Slave Features:

|                                  |          |
|----------------------------------|----------|
| Clock No Stretch Mode            | Disabled |
| General Call Address Detection   | Disabled |
| Primary Address Length selection | 7-bit    |
| Dual Address Acknowledged        | Disabled |
| Primary slave address            | 0        |

### 3.3. NUCLEO-L476RG

#### mode: Human Machine Interface

##### 3.3.1. Human Machine Interface:

###### Led:

|                       |               |
|-----------------------|---------------|
| USER LED GREEN (LED2) | <b>true *</b> |
|-----------------------|---------------|

###### Button:

|             |                    |
|-------------|--------------------|
| USER BUTTON | <b>Mode EXTI *</b> |
|-------------|--------------------|

###### Demonstration code:

|                             |          |
|-----------------------------|----------|
| Generate demonstration code | Disabled |
|-----------------------------|----------|

### 3.4. RCC

##### 3.4.1. Parameter Settings:

###### System Parameters:

|                 |     |
|-----------------|-----|
| VDD voltage (V) | 3.3 |
|-----------------|-----|

|                                |                                 |
|--------------------------------|---------------------------------|
| Instruction Cache              | Enabled                         |
| Prefetch Buffer                | <b>Enabled *</b>                |
| Data Cache                     | Enabled                         |
| Flash Latency(WS)              | 4 WS (5 CPU cycle)              |
| <b>RCC Parameters:</b>         |                                 |
| HSI Calibration Value          | 16                              |
| MSI Calibration Value          | 0                               |
| MSI Auto Calibration           | Disabled                        |
| HSE Startup Timeout Value (ms) | 100                             |
| LSE Startup Timeout Value (ms) | 5000                            |
| <b>Power Parameters:</b>       |                                 |
| Power Regulator Voltage Scale  | Power Regulator Voltage Scale 1 |

### 3.5. SYS

#### Debug: Serial Wire

#### Timebase Source: TIM6

### 3.6. USART1

#### Mode: Asynchronous

#### 3.6.1. Parameter Settings:

##### Basic Parameters:

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

##### Advanced Parameters:

|                |                      |
|----------------|----------------------|
| Data Direction | Receive and Transmit |
| Over Sampling  | 16 Samples           |
| Single Sample  | Disable              |

##### Advanced Features:

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.7. FREERTOS

#### Interface: CMSIS\_V2

##### 3.7.1. Config parameters:

###### API:

FreeRTOS API CMSIS v2

###### Versions:

FreeRTOS version 10.3.1

CMSIS-RTOS version 2.1.3

###### MPU/FPU:

ENABLE\_MPU Disabled

ENABLE\_FPU Disabled

###### Kernel settings:

USE\_PREEMPTION Enabled

CPU\_CLOCK\_HZ SystemCoreClock

TICK\_RATE\_HZ 1000

MAX\_PRIORITIES 56

USE\_SB\_COMPLETED\_CALLBACK 0

USE\_MINI\_LIST\_ITEM 1

MINIMAL\_STACK\_SIZE 128

MAX\_TASK\_NAME\_LEN 16

USE\_16\_BIT\_TICKS Disabled

IDLE\_SHOULD\_YIELD Enabled

USE\_MUTEXES Enabled

USE\_RECURSIVE\_MUTEXES Enabled

USE\_COUNTING\_SEMAPHORES Enabled

QUEUE\_REGISTRY\_SIZE 8

USE\_APPLICATION\_TASK\_TAG Disabled

ENABLE\_BACKWARD\_COMPATIBILITY Enabled

USE\_PORT\_OPTIMISED\_TASK\_SELECTION Disabled

USE\_TICKLESS\_IDLE Disabled

USE\_TASK\_NOTIFICATIONS Enabled

RECORD\_STACK\_HIGH\_ADDRESS Disabled

###### Memory management settings:

Memory Allocation Dynamic / Static

TOTAL\_HEAP\_SIZE **8192 \***

Memory Management scheme heap\_4

###### Hook function related definitions:

|                              |          |
|------------------------------|----------|
| USE_IDLE_HOOK                | Disabled |
| USE_TICK_HOOK                | Disabled |
| USE_MALLOC_FAILED_HOOK       | Disabled |
| USE_DAEMON_TASK_STARTUP_HOOK | Disabled |
| CHECK_FOR_STACK_OVERFLOW     | Disabled |

**Run time and task stats gathering related definitions:**

|                                |          |
|--------------------------------|----------|
| GENERATE_RUN_TIME_STATS        | Disabled |
| USE_TRACE_FACILITY             | Enabled  |
| USE_STATS_FORMATTING_FUNCTIONS | Disabled |

**Co-routine related definitions:**

|                           |          |
|---------------------------|----------|
| USE_CO_ROUTINES           | Disabled |
| MAX_CO_ROUTINE_PRIORITIES | 2        |

**Software timer definitions:**

|                        |         |
|------------------------|---------|
| USE_TIMERS             | Enabled |
| TIMER_TASK_PRIORITY    | 2       |
| TIMER_QUEUE_LENGTH     | 10      |
| TIMER_TASK_STACK_DEPTH | 256     |

**Interrupt nesting behaviour configuration:**

|                                        |    |
|----------------------------------------|----|
| LIBRARY_LOWEST_INTERRUPT_PRIORITY      | 15 |
| LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY | 5  |

**Added with 10.2.1 support:**

|                            |          |
|----------------------------|----------|
| MESSAGE_BUFFER_LENGTH_TYPE | size_t   |
| USE_POSIX_ERRNO            | Disabled |

**CMSIS-RTOS V2 flags:**

|                               |         |
|-------------------------------|---------|
| USE_OS2_THREAD_SUSPEND_RESUME | Enabled |
| USE_OS2_THREAD_ENUMERATE      | Enabled |
| USE_OS2_EVENTFLAGS_FROM_ISR   | Enabled |
| USE_OS2_THREAD_FLAGS          | Enabled |
| USE_OS2_TIMER                 | Enabled |
| USE_OS2_MUTEX                 | Enabled |

**3.7.2. Include parameters:**

**Include definitions:**

|                       |          |
|-----------------------|----------|
| vTaskPrioritySet      | Enabled  |
| uxTaskPriorityGet     | Enabled  |
| vTaskDelete           | Enabled  |
| vTaskCleanUpResources | Disabled |
| vTaskSuspend          | Enabled  |
| vTaskDelayUntil       | Enabled  |
| vTaskDelay            | Enabled  |

|                              |          |
|------------------------------|----------|
| xTaskGetSchedulerState       | Enabled  |
| xTaskResumeFromISR           | Enabled  |
| xQueueGetMutexHolder         | Enabled  |
| xSemaphoreGetMutexHolder     | Disabled |
| pcTaskGetTaskName            | Disabled |
| uxTaskGetStackHighWaterMark  | Enabled  |
| xTaskGetCurrentTaskHandle    | Enabled  |
| eTaskGetState                | Enabled  |
| xEventGroupSetBitFromISR     | Disabled |
| xTimerPendFunctionCall       | Enabled  |
| xTaskAbortDelay              | Disabled |
| xTaskGetHandle               | Disabled |
| uxTaskGetStackHighWaterMark2 | Disabled |

### 3.7.3. Advanced settings:

#### **Newlib settings (see parameter description first):**

|                      |          |
|----------------------|----------|
| USE_NEWLIB_REENTRANT | Disabled |
|----------------------|----------|

#### **Project settings (see parameter description first):**

|                       |         |
|-----------------------|---------|
| Use FW pack heap file | Enabled |
|-----------------------|---------|

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP                    | Pin                   | Signal         | GPIO mode                      | GPIO pull/up pull down      | Max Speed   | User Label      |
|-----------------------|-----------------------|----------------|--------------------------------|-----------------------------|-------------|-----------------|
| ADC1                  | PC0                   | ADC1_IN1       | Analog mode for ADC conversion | No pull-up and no pull-down | n/a         | NTC_AD1         |
| I2C1                  | PB8                   | I2C1_SCL       | Alternate Function Open Drain  | No pull-up and no pull-down | Very High * |                 |
|                       | PB9                   | I2C1_SDA       | Alternate Function Open Drain  | No pull-up and no pull-down | Very High * |                 |
| SYS                   | PA13 (JTMS-SWDIO)     | SYS_JTMS-SWDIO | n/a                            | n/a                         | n/a         | TMS             |
|                       | PA14 (JTCK-SWCLK)     | SYS_JTCK-SWCLK | n/a                            | n/a                         | n/a         | TCK             |
| USART1                | PA9                   | USART1_TX      | Alternate Function Push Pull   | No pull-up and no pull-down | Very High * |                 |
|                       | PA10                  | USART1_RX      | Alternate Function Push Pull   | No pull-up and no pull-down | Very High * |                 |
| Single Mapped Signals | PC14-OSC32_IN (PC14)  | RCC_OSC32_IN   | n/a                            | n/a                         | n/a         |                 |
|                       | PC15-OSC32_OUT (PC15) | RCC_OSC32_OUT  | n/a                            | n/a                         | n/a         |                 |
|                       | PH0-OSC_IN (PH0)      | RCC_OSC_IN     | n/a                            | n/a                         | n/a         |                 |
|                       | PH1-OSC_OUT (PH1)     | RCC_OSC_OUT    | n/a                            | n/a                         | n/a         |                 |
|                       | PB3 (JTDO-TRACESW)    | SYS_JTDO-SWO   | n/a                            | n/a                         | n/a         | SWO             |
| GPIO                  | PA0                   | GPIO_Output    | Output Push Pull               | No pull-up and no pull-down | Low         | HeatRelay       |
|                       | PA1                   | GPIO_Output    | Output Push Pull               | No pull-up and no pull-down | Low         | FanRelay        |
|                       | PA4                   | GPIO_Output    | Output Push Pull               | No pull-up and no pull-down | Low         | HumidRelay      |
|                       | PC5                   | GPIO_Output    | Output Push Pull               | No pull-up and no pull-down | Low         | Buzzer          |
|                       | PC6                   | GPIO_Input     | Input mode                     | No pull-up and no pull-down | n/a         | UpArrowSwitch   |
|                       | PC7                   | GPIO_Input     | Input mode                     | No pull-up and no pull-down | n/a         | DownArrowSwitch |
|                       | PC8                   | GPIO_Input     | Input mode                     | No pull-up and no pull-down | n/a         | ModeSwitch      |
|                       | PC9                   | GPIO_Input     | Input mode                     | No pull-up and no pull-down | n/a         | EnterSwitch     |

| IP | Pin | Signal      | GPIO mode        | GPIO pull/up pull down      | Max Speed | User Label   |
|----|-----|-------------|------------------|-----------------------------|-----------|--------------|
|    | PA8 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low       | OnOffHeatLed |

#### 4.2. DMA configuration

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                                                            | Enable | Preenmption Priority | SubPriority |
|----------------------------------------------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                                                     | true   | 0                    | 0           |
| Hard fault interrupt                                                       | true   | 0                    | 0           |
| Memory management fault                                                    | true   | 0                    | 0           |
| Prefetch fault, memory access fault                                        | true   | 0                    | 0           |
| Undefined instruction or illegal state                                     | true   | 0                    | 0           |
| System service call via SWI instruction                                    | true   | 0                    | 0           |
| Debug monitor                                                              | true   | 0                    | 0           |
| Pendable request for system service                                        | true   | 15                   | 0           |
| System tick timer                                                          | true   | 15                   | 0           |
| USART1 global interrupt                                                    | true   | 5                    | 0           |
| EXTI line[15:10] interrupts                                                | true   | 5                    | 0           |
| TIM6 global interrupt, DAC channel1 and channel2 underrun error interrupts | true   | 15                   | 0           |
| PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/35/36/37/38       | unused |                      |             |
| Flash global interrupt                                                     | unused |                      |             |
| RCC global interrupt                                                       | unused |                      |             |
| ADC1 and ADC2 interrupts                                                   | unused |                      |             |
| I2C1 event interrupt                                                       | unused |                      |             |
| I2C1 error interrupt                                                       | unused |                      |             |
| FPU global interrupt                                                       | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                                                    | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|----------------------------------------------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                                                     | false                             | true                 | false            |
| Hard fault interrupt                                                       | false                             | true                 | false            |
| Memory management fault                                                    | false                             | true                 | false            |
| Prefetch fault, memory access fault                                        | false                             | true                 | false            |
| Undefined instruction or illegal state                                     | false                             | true                 | false            |
| System service call via SWI instruction                                    | false                             | false                | false            |
| Debug monitor                                                              | false                             | true                 | false            |
| Pendable request for system service                                        | false                             | false                | false            |
| System tick timer                                                          | false                             | false                | true             |
| USART1 global interrupt                                                    | false                             | true                 | true             |
| EXTI line[15:10] interrupts                                                | false                             | true                 | true             |
| TIM6 global interrupt, DAC channel1 and channel2 underrun error interrupts | false                             | true                 | true             |

**\* User modified value**

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

Middleware

FREERTOS

| System Core | Analog | Timers | Connectivity | Multimedia | Security | Computing | Bsp           |
|-------------|--------|--------|--------------|------------|----------|-----------|---------------|
| DMA         | ADC1   |        | I2C1         |            |          |           | NUCLEO-L476RG |
| GPIO        |        |        | USART1       |            |          |           |               |
| IIVIC       |        |        |              |            |          |           |               |
| RCC         |        |        |              |            |          |           |               |
| SYS         |        |        |              |            |          |           |               |

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32l4_bsd.zip">https://www.st.com/resource/en/bsdl_model/stm32l4_bsd.zip</a>                                                                                                                                                           |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32l4_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32l4_ibis.zip</a>                                                                                                                                                         |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32l4_svd.zip">https://www.st.com/resource/en/svd/stm32l4_svd.zip</a>                                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                   |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                   |
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| Application Notes   | <a href="https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-for-stm32-mcus-stmicroelectronics.pdf</a> |
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| Errata Sheets              | <a href="https://www.st.com/resource/en/errata_sheet/es0250-stm32l471xx475xx476xx486xx-device-errata-stmicroelectronics.pdf">https://www.st.com/resource/en/errata_sheet/es0250-stm32l471xx475xx476xx486xx-device-errata-stmicroelectronics.pdf</a>                                                                                                           |
| Datasheet                  | <a href="https://www.st.com/resource/en/datasheet/dm00108832.pdf">https://www.st.com/resource/en/datasheet/dm00108832.pdf</a>                                                                                                                                                                                                                                 |
| Programming Manuals        | <a href="https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                                                                 |
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| User Manuals               | <a href="https://www.st.com/resource/en/user_manual/um3166-stm32l4-and-">https://www.st.com/resource/en/user_manual/um3166-stm32l4-and-</a>                                                                                                                                                                                                                   |

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